



Solutions:

1. C. $4.0 \times 10^{-4} \text{ N}$

Solution:

Arrange the charges in the xy plane at locations (4,4), (4,-4), (-4,4), and (-4,-4). Then the fifth charge will be on the z axis at location $z = 4\sqrt{2}$, which puts it at 8cm distance from the other four. By symmetry, the force on the fifth charge will be z-directed, and will be four times the z component of force produced by each of the four other charges.

$$F = \frac{4}{\sqrt{2}} \left(\frac{q^2}{4\pi\epsilon_0 d^2} \right)$$

$$F = \frac{4}{\sqrt{2}} \left[\frac{(10^{-8})^2}{4\pi(8.85 \times 10^{-12})(0.08)^2} \right]$$

$$F = 4.0 \times 10^{-4} \text{ N}$$

2. B. $21.5a_x \mu\text{N}$

Solution:

$$F = \frac{(50 \times 10^{-9})^2}{4\pi\epsilon_0} \left[\frac{R_{CA}}{|R_{CA}|^3} + \frac{R_{DA}}{|R_{DA}|^3} + \frac{R_{BA}}{|R_{BA}|^3} \right]$$

Where:

$$R_{CA} = a_x - a_y$$

$$R_{DA} = a_x + a_y$$

$$R_{BA} = 2a_x$$

The magnitudes are:

$$|R_{CA}| = |R_{DA}| = \sqrt{2}$$

$$|R_{BA}| = 2$$

Substituting:

$$F = \frac{(50 \times 10^{-9})^2}{4\pi\epsilon_0} \left[\frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} + \frac{2}{8} \right] a_x$$

$$F = 21.5a_x \mu\text{N}$$

3. B. -43.3

Solution:

The finite probability effectively reduces the net charge quantity by the probability fraction. With $e = -1.602 \times 10^{-19} \text{ C}$, the density becomes:

$$\rho_v = -\frac{0.27 \times 1.602 \times 10^{-19}}{10^{-15}}$$

$$\rho_v = -43.3 \mu\text{C/m}^3$$

4. B. 1.93 N directed from q to Q

Solution:

$$|F_Q| = \frac{|q||Q|}{4\pi\epsilon_0 R^2}$$

$$= \frac{(300 \times 10^{-6})(10 \times 10^{-6})}{4\pi \left(\frac{10^{-9}}{36\pi} \right) [(1-2)^2 + (1-4)^2]}$$

$$= 1.93 \text{ N}$$

The force is directed from q toward Q.

5. D. 18

Solution:

$$(\text{curl } E)_x = (\nabla \times E)_x$$

$$= \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}$$

$$= xz - (-2zx)$$

$$= 3xz$$

$$= 3(2)(3)$$

$$= 18$$

6. D. $\nabla(\nabla \cdot \mathbf{P}) - \nabla^2 \mathbf{P}$

7. D. $\nabla \times \mathbf{E} = 0$ and $\nabla \cdot \mathbf{B} = 0$

8. A. whose curl is equal to the magnetic flux density

9. C. 1

Solution:

$$\text{div } \vec{A} = \nabla \cdot \vec{A}$$

$$= \left(\frac{\partial i}{\partial x} + \frac{\partial i}{\partial y} + \frac{\partial k}{\partial z} \right) \cdot \begin{pmatrix} x\hat{a}_x + y\hat{a}_y \\ + z\hat{a}_z \end{pmatrix}$$

$$= \left(\frac{\partial i}{\partial x} + \frac{\partial i}{\partial y} + \frac{\partial k}{\partial z} \right) \cdot (xi + yi + zk)$$

$$= 1 + 1 + 1$$

$$= 3$$

Note: $i \cdot i = i \cdot i = k \cdot k = 1$

10. D. x, z components

11. C. $1/120\pi$

Solution:

Time average power density:

$$P_{av} = \frac{1}{2} EH$$

$$= \frac{1}{2} \times \frac{E^2}{\eta}$$

Intrinsic impedance of EM wave:

$$\gamma = \sqrt{\frac{\mu}{\epsilon}}$$

$$\eta = \sqrt{\frac{\mu_0}{4\epsilon_0}}$$

$$\eta = \frac{120\pi}{2}$$

$$\eta = 60\pi$$

$$P_{av} = \frac{E^2}{2\eta}$$

$$P_{av} = \frac{1}{2(60\pi)}$$

$$P_{av} = \frac{1}{120\pi}$$

12. A. Zero

Solution:

The point charge +q will induce a charge -q on the surface of metal sheet sphere. Using Gauss's law, the net electric flux passing through a closed surface is equal to the charge enclosed = +q - q = 0
D = 0, E = 0 at point P.



13. A. ρ

Solution:

$$\begin{aligned}\text{div } D &= \lim_{V \rightarrow 0} \frac{\int_V D \cdot dS}{V} \\ &= \lim_{V \rightarrow 0} \left(\frac{1}{V} \int_V \rho dV \right) \\ &= \rho\end{aligned}$$

14. A. 40π coulombs

Solution:

The sphere is enclosed in a cube of side = 10m. using Gauss's law, the net electric flux flowing out through a closed surface is equal to charge enclosed.

$$\begin{aligned}\oint_S \vec{D} \cdot d\vec{a} &= Q \text{ (enclosed)} \\ &= P_s (4\pi r^2) \\ &= 10(4\pi) \\ &= 40\pi \text{ C}\end{aligned}$$

15. A. $q = (-4/9)Q$, $x = (2/3)d$

Solution:

The unknown charge q has to be negative and located between Q and $4Q$. For equilibrium, $F_{12} = F_{32} = F_{13}$. Hence,

$$\frac{Qq}{(d-x)^2} = \frac{4Qq}{x^2} = \frac{(4Q)(Q)}{d^2}$$

Solving above equation yields:

$$\begin{aligned}q &= -\frac{4}{9}Q \\ x &= \frac{2}{3}d\end{aligned}$$

16. B. Minimum

17. B. 6

Solution:

$$\text{div } E = \nabla \cdot E$$

$$\begin{aligned}&= \left[\frac{\partial}{\partial x}(xy^2) + \frac{\partial}{\partial y}(-xz^2) \right] \\ &\quad + \frac{\partial}{\partial z}(xyz) \\ &= y^2 + xy \\ &= 2^2 + (1)(2) \\ &= 6\end{aligned}$$

18. B. 0.72 N

Solution:

$$\begin{aligned}|F| &= \frac{(100 \times 10^{-6})(20 \times 10^{-6})}{4\pi \left(\frac{10^{-9}}{36\pi} \right) (4^2 + 0^2 + 3^2)} \\ &= 0.72 \text{ N}\end{aligned}$$

19. D. $3/\sqrt{19}$, $-1/\sqrt{19}$ and $-3/\sqrt{19}$

Solution:

The force, which is attractive, is parallel to the vector $3a_x - 1a_y - 3a_z$. Hence, the desired direction cosines are:

$$\begin{aligned}\alpha &= \frac{3}{\sqrt{19}} \\ \beta &= \frac{-1}{\sqrt{19}} \\ \gamma &= \frac{-3}{\sqrt{19}}\end{aligned}$$

20. A. $1.718a_x + 0.859a_y + 2.577a_z$

Solution:

$$\begin{aligned}F_{21} &= \frac{Q_1 Q_2}{4\pi \epsilon_0 R_{21}^3} R_{21} \\ &= \left[\frac{(100 \times 10^{-6})(50 \times 10^{-6})}{\left(\frac{10^{-9}}{9} \right) (2^2 + 1^2 + 3^2)^{3/2}} \right] \\ &\quad (2a_x + a_y + 3a_z) \\ &= 1.718a_x + 0.859a_y + 2.577a_z \text{ N}\end{aligned}$$

21. C. At infinity

22. C. 0.789 C

Solution:

$$\begin{aligned}Q &= \int_V \rho_v dV \\ &= \int_{r=0}^{0.200} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \left(\frac{2}{r \sin \theta} \right) r^2 \sin \theta dr d\theta d\phi \\ &= \int_0^{0.200} 2r dr \int_0^{\pi} d\theta \int_0^{2\pi} d\phi \\ &= 0.08\pi^2 \\ &Q \approx 0.789 \text{ C}\end{aligned}$$

23. C. -30 C

Solution:

Representing the line segment by:

$$\begin{aligned}x &= 2 + t(4-2) \\ y &= 1 + t(3-1) \\ z &= 5 + t(6-5)\end{aligned}$$

with $0 \leq t \leq 1$, one has $\rho(t) = -13 + 6t$ and

$$\begin{aligned}ds &= \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 + \left(\frac{dz}{dt} \right)^2} dt \\ &= \sqrt{2^2 + 2^2 + 1^2} dt = 3dt \\ Q &= \int_0^1 (-13 + 6t)(3dt) \\ &= -30 \text{ C}\end{aligned}$$

24. C. Product of charges

25. A. $(1/3)2\pi kR^3$

$$\begin{aligned}\int_S \rho_s ds &= \int_0^R (kr)(2\pi r) dr \\ &= \frac{2\pi kR^3}{3}\end{aligned}$$

END